

Ludwig Adriane G. Intal

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SUMMARY

A Computer Engineer experienced in integrating software and hardware to develop functional systems with the help of AI. Skilled in building and refining working programs while transforming messy and unstructured data into clean, organized, and readable datasets for analysis and decision-making.

TECHNICAL SKILLS

- **Tech Stack:** C++, Python, Flutter, MySQL, PostgreSQL
- **Hardware:** ESP32, Raspberry Pi 4B, Arduino UNO, KiCad
- **Skills:** Excel, PowerBI, Artificial Intelligence, Computer Vision, Prompt Engineering
- **Systems and Tools:** Windows, Linux, Git, Github
- **AI:** Claude, Gemini, ChatGPT, DeepSeek

WORK EXPERIENCE

Municipality of San Simon Pampanga

February 2026 - April 2026

Internship

- Installed and configured linux operating system on kiosk.
- Applied Excel formulas to automate data entry, remove duplicate records, perform calculations, and compare datasets, improving data accuracy and processing efficiency.

PROJECTS

WetWare — Real-Time Water Quality Monitoring System

Software Leader | ESP32 (C++) | JavaScript (ES6) | LittleFS & HTML5 Canvas

- Engineered an ESP32-based IoT water monitoring system with real-time pH, TDS, and temperature monitoring, a responsive web dashboard with secure controls, and persistent CSV data logging using ESP32 LittleFS storage for graph visualization and reporting.

BraiLingo — Braille Learning Device for Blind and Visually Impaired Users

Software Leader | Raspberry Pi 4B + Flutter Android App | Undergraduate Thesis

- Developed a Raspberry Pi 4B-powered tactile Braille learning device using PWM-controlled solenoid actuators driven through a PCF8575 I2C GPIO expander, integrated with a companion Flutter mobile app over TCP to deliver adaptive, spaced-repetition Braille quizzes.

EDUCATION

Bachelor of Science in Computer Engineering

2022-2026

Pampanga State University (former Don Honorio Ventura State University)

CERTIFICATES

- **"Data Analytics Essentials"** offered by DICT-ITU DTC Initiative through the Cisco Networking Academy program
- **How Data Analytics Powers Strategic Decision-Making: Leveraging Power BI, SQL, Data Warehousing, and Python** Institute of Computer Engineering of the Philippines Region 3